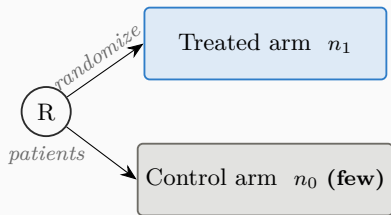


Randomized trial (small, valid)



covariate shift
different case-mix X

External control source (large)

compatible incompatible



a hidden subset is incompatible (shifted control outcome)

keep compatible

Residual-rank screen
rank each EC among the RCT controls

× drop incompatible

$$\hat{\tau} = \bar{Y}_{\text{treated}} - \bar{Y}_{\text{RCT controls} + \text{borrowed compatible ECs}}$$

more valid controls $\Rightarrow$ sharper $\hat{\tau}$ — only if the borrowed ones are truly compatible